Homework 1

EN.601.639 | Fall 2026

You must submit either a physical handwritten homework or print out your homework to submit at the beginning of class.

Problem 1: PCA for Different Distributions

a) Write out the PMF of a negative binomial distribution. When would you use a negative binomial distribution versus a Poisson distribution?

b) Write out the GLM-PCA optimization problem for single cell count data following a negative binomial distribution. (Hint: The negative binomial distribution has two arguments, mean and overdispersion. Only use the matrix factorization UV for one of these two arguments, and set the other as a learned constant.)

c) What is the PDF of a log-normal distribution? When would you use such a distribution?

d) Suppose your collaborator generated metabolomics data and wanted you to run GLM-PCA on it. After preprocessing, metabolomics data is often approximated with a log-normal distribution. Write out the GLM-PCA optimization problem for this metabolomics data.

e) Suppose you asked ChatGPT to answer Question 1d for a brand new, very complicated distribution, and it gave you a specific optimization problem. However, you did not trust ChatGPT's answer and wanted to double-check that it gave you the correct optimization problem. Suggest a way you could do so without re-deriving the optimization problem from scratch.

Problem 2: Canonical correlation analysis (CCA)

We are given two single cell datasets

$$X = [x_{ig}] \in \mathbb{R}^{N\times G} \qquad \text{and} \qquad Y = [y_{ip}] \in \mathbb{R}^{N\times P}, \tag{1}$$

where each dataset measures a molecular feature (e.g., gene expression, protein expression, chromatin accessibility) on N cells. The two datasets measure different features, with sizes G and P respectively, on the same N cells. Assume X and Y are mean-centered, meaning that $\frac{1}{N}\sum_{i=1}^{N} x_{ig} = 0$ and $\frac{1}{N}\sum_{i=1}^{N} y_{ip} = 0$ for all features $g = 1, \ldots, G$ and $p = 1, \ldots, P$.

Canonical correlation analysis (CCA) is an approach to identify "shared correlation structures" for the two datasets X and Y. Specifically, in CCA, one aims to identify linear combinations

$$u \in \mathbb{R}^{G} \qquad \text{and} \qquad v \in \mathbb{R}^{P}, \tag{2}$$

of the two features such that Xu and Yv are maximally correlated.

a) Fill in the blanks in the CCA optimization problem:

$$\underset{\substack{||u||_2=1 \\ ||v||_2=1}}{\operatorname{argmax}} \ \operatorname{Corr}\left(\underline{\hspace{2cm}}, \underline{\hspace{2cm}}\right) \tag{3}$$

b) Suppose $X = Y$ and we replace Corr with Cov in the optimization problem in part a. What problem does this represent? (Hint: We spent a class on it.)

c) Recall the formula for Pearson correlation between two vectors a and b:

$$\operatorname{Corr}(a, b) = \frac{\operatorname{Cov}(a, b)}{\sqrt{\operatorname{Var}(a)\operatorname{Var}(b)}} \tag{4}$$

Rewrite the optimization problem in part **a** using this equation.

d) Define the variables $\Sigma_{XY} = \frac{1}{N-1}X^TY$, $\Sigma_X = \frac{1}{N-1}X^TX$, and $\Sigma_Y = \frac{1}{N-1}Y^TY$.

Rewrite the optimization problem in part **c** to remove the variables X and Y, and instead use the variables Σ_{XY}, Σ_X, and Σ_Y.

The variable Σ_{XY} is called the *sample cross-covariance matrix* and the variables Σ_X and Σ_Y are called the *sample covariance matrices* for X and Y, respectively.

Hint: Your optimization problem should have the form

$$\underset{||v||_2=1, ||w||_2=1}{\operatorname{argmax}} \frac{\underline{\hspace{4cm}}}{\sqrt{\underline{\hspace{2cm}}}\sqrt{\underline{\hspace{2cm}}}} \tag{5}$$

e) Define the variables

$$\widetilde{u} = \Sigma_X^{1/2} u \qquad \text{and} \qquad \widetilde{v} = \Sigma_Y^{1/2} v. \tag{6}$$

Rewrite the optimization problem in part **d** in terms of the variables $\widetilde{u}, \widetilde{v}, \Sigma_X, \Sigma_Y, \Sigma_{XY}$.

f) Suppose you are told that a solution to the optimization problem

$$\underset{\substack{||a||_2=1\\ ||b||_2=1}}{\operatorname{argmax}}\, a^T M b \tag{7}$$

is given by

$$\widehat{a} = u \qquad \text{and} \qquad \widehat{b} = v, \tag{8}$$

where u and v are the (matched) leading left and right singular vectors of the matrix M.

Use this knowledge to derive the solution to the optimization problem in part **e**?

g) *[Open-ended]* The procedure you derived gives you the first *canonical correlation vectors* u_1 and v_1. How could you extend the approach in part **f** to find the second canonical correlation vectors u_2 and v_2? More generally, how could you find the k-th canonical correlation vectors u_k and v_k for any k?

h) Consider a different setting where we are given two single cell datasets

$$X \in \mathbb{R}^{N\times G} \qquad \text{and} \qquad Y \in \mathbb{R}^{M\times G} \tag{9}$$

which measure the same molecular feature (e.g., gene expression) on two cell populations of sizes N and M, respectively.

We would like to use CCA to embed the N cells and M cells, respectively, into a shared embedding space. How could you use CCA to do this?

[CCA is the core algorithmic idea behind the first version of Seurat, a very popular single cell analysis package.]

Problem 3: NMF for multiple samples and modalities

a) Suppose we measure a single cell data matrix $X \in \mathbb{R}^{N \times P}$ for N cells with P features. We aim to factorize the observed data as $X \approx UV$, where $U \in \mathbb{R}^{N \times K}$ is the factor matrix and $V \in \mathbb{K} \times \mathbb{G}$ is the loading matrix, for some small $K > 0$. Write out the NMF optimization problem for computing this factorization.

b) Next, suppose we measure S different single cell data matrices $X^{(s)} \in \mathbb{R}^{N_s \times P}$ for $s = 1, \ldots, S$. Our data measures the same P features across S different *samples*, i.e. populations of cells. We aim to factorize each sample as $X^{(s)} \approx U^{(s)}V$, where each sample s has its own factor matrix $U^{(s)}$ but all samples share a loading matrix V.

Write out an NMF-style optimization problem for computing this factorization. Qualitatively explain what biological assumption is made if we share the loading matrix V across all samples.

c) Next, suppose we measure M different single cell data matrices $X^{(m)} \in \mathbb{R}^{N \times P_m}$ for $m = 1, \ldots, M$. Our data measures M different molecular modalities (e.g. gene expression, chromatin accessibility, etc.) across the same N cells. We aim to factorize each sample as $X^{(m)} \approx UV^{(m)}$, where each modality m has its own loading matrix $V^{(m)}$ but all modalities share a factor matrix U.

Write out an NMF-style optimization problem for computing this factorization. Qualitatively explain what biological assumption is made if we share the factor matrix U across all samples.

d) Next, suppose we measure $S \cdot M$ different single cell data matrices $X^{(s,m)} \in \mathbb{R}^{N_s \times P_m}$ for $s = 1, \ldots, S$ and $m = 1, \ldots, M$. Our data measures M different molecular modalities across S different samples. We aim to factorize each sample as $X^{(s,m)} \approx U^{(s)}V^{(m)}$, where each sample s has its own factor matrix $U^{(s)}$ and each modality m has its own loading matrix $V^{(m)}$.

Write out an NMF-style optimization problem for computing this factorization. Qualitatively explain what biological assumption(s) are being made by this factorization.

e) In practice, not every modality may be measured in every sample. Let $\Omega \subseteq \{1, \ldots, S\} \times \{1, \ldots, M\}$ be the set of observed pairs (s, m) of samples and modalities. sample-modality pairs. We only observe single cell data matrices $X^{(s,m)}$ for sample-modality pairs $(s, m) \in \Omega$.

Modify the optimization problem in part **d** to only incorporate the observed data.

f) Assume that observed pairs are sufficient to learn every factor matrix $U^{(s)}$ and loading matrix $V^{(m)}$. How would you estimate an *unobserved* $X^{(s,m)}$ (i.e. $(s, m) \notin \Omega$)?